

EDA Code

```
# Import plotting library
library(ggplot2)

# Reads in data frame
df <- read.csv("insurance.csv")

# Sets categorical variables as factors
df$children <- as.factor(df$children)
df$sex <- as.factor(df$sex)
df$smoker <- as.factor(df$smoker)
df$region <- as.factor(df$region)

# Prints head of data frame
head(df)
```

```
##   age    sex    bmi children smoker   region   charges
## 1  19 female 27.900         0    yes southwest 16884.924
## 2  18  male 33.770         1    no  southeast  1725.552
## 3  28  male 33.000         3    no  southeast  4449.462
## 4  33  male 22.705         0    no northwest 21984.471
## 5  32  male 28.880         0    no northwest  3866.855
## 6  31 female 25.740         0    no  southeast  3756.622
```

```
# Checks for any missing values in the data set
sum(is.na(df))
```

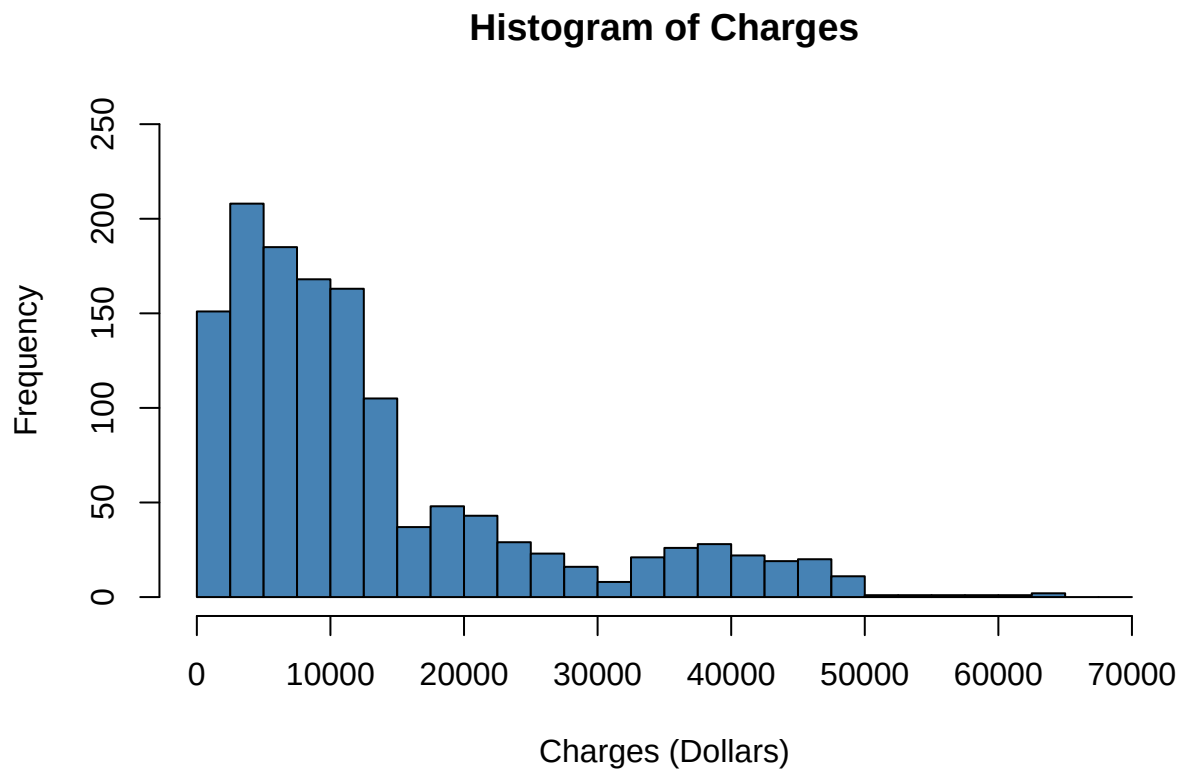
```
## [1] 0
```

```
# Gives summary statistics for each variable in the data set
summary(df)
```

```
##      age      sex      bmi      children smoker
##  Min.   :18.00  female:662  Min.   :15.96  0:574    no :1064
## 1st Qu.:27.00  male  :676  1st Qu.:26.30  1:324    yes: 274
## Median :39.00
## Mean   :39.21
## 3rd Qu.:51.00
## Max.   :64.00
##      region      charges
## northeast:324  Min.   : 1122
## northwest:325 1st Qu.: 4740
## southeast:364 Median : 9382
## southwest:325 Mean    :13270
##              3rd Qu.:16640
##              Max.    :63770
```

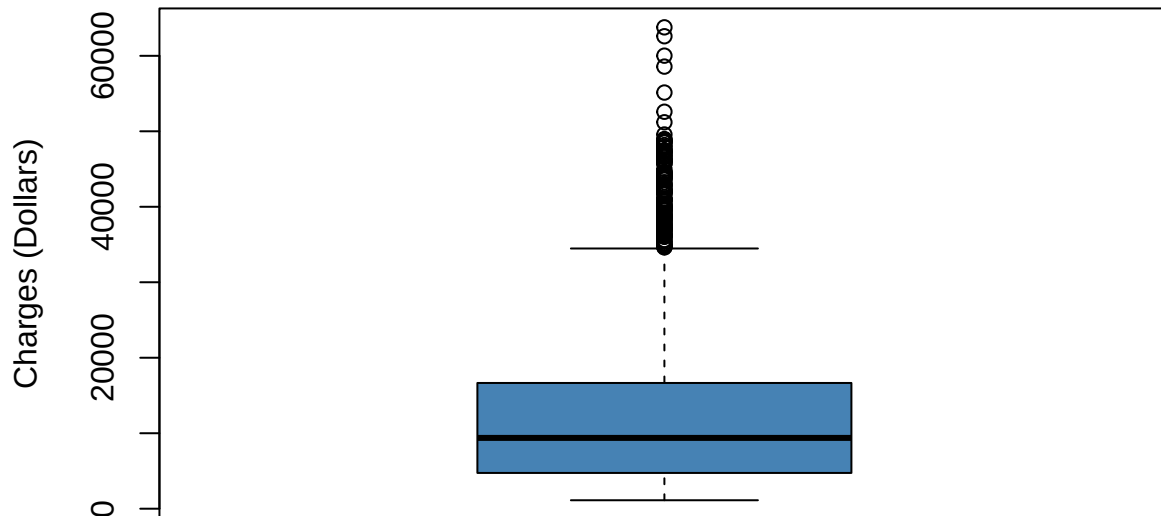
```
# Makes breaks for histogram
hist_breaks <- seq(0, 70000, by = 2500)

# Creates histogram of insurance charges
hist(df$charges, main="Histogram of Charges", xlab="Charges (Dollars)",
      xlim=range(0,70000), breaks=hist_breaks, ylim=range(0,250),
      col="steelblue")
```



```
# Creates box plot of charges
boxplot(df$charges, main="Box Plot of Charges", ylab="Charges (Dollars)",
        col="steelblue")
```

Box Plot of Charges



```
# Sets x equal to charges
x <- df$charges

# Finds the first and third quartile of charges
Q1 <- quantile(x, 0.25)
Q3 <- quantile(x, 0.75)

# Gets the interquartile range
IQR_val <- IQR(x)

# Finds the lower and upper bounds
lower_bound <- Q1 - (1.5 * IQR_val)
upper_bound <- Q3 + (1.5 * IQR_val)

# Counts outliers
sum(x > upper_bound)
```

```
## [1] 139
```

```
# Creates log transformation of charges
df$log_charges <- log(df$charges)

# Gives summary statistics for new variable
summary(df$log_charges)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
```

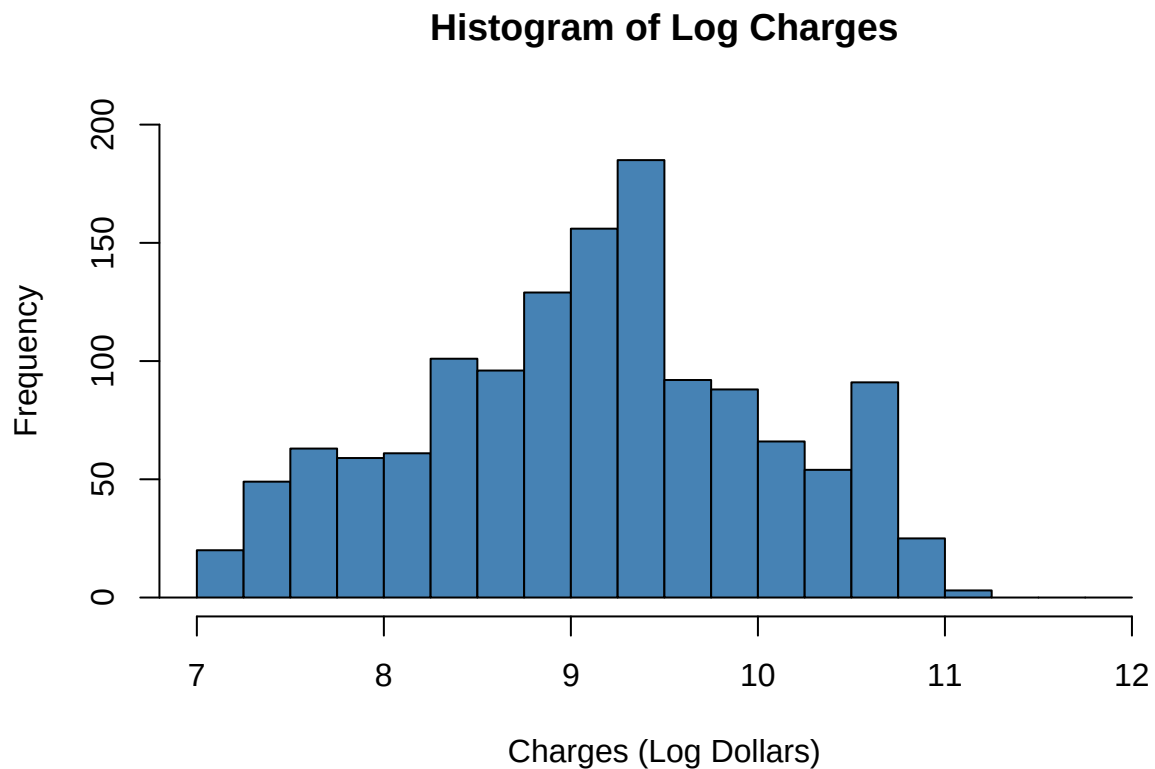
```
## 7.023 8.464 9.147 9.099 9.720 11.063
```

```
# Makes breaks for histogram
```

```
hist_breaks <- seq(0, 12, by = 0.25)
```

```
# Creates histogram of log insurance charges
```

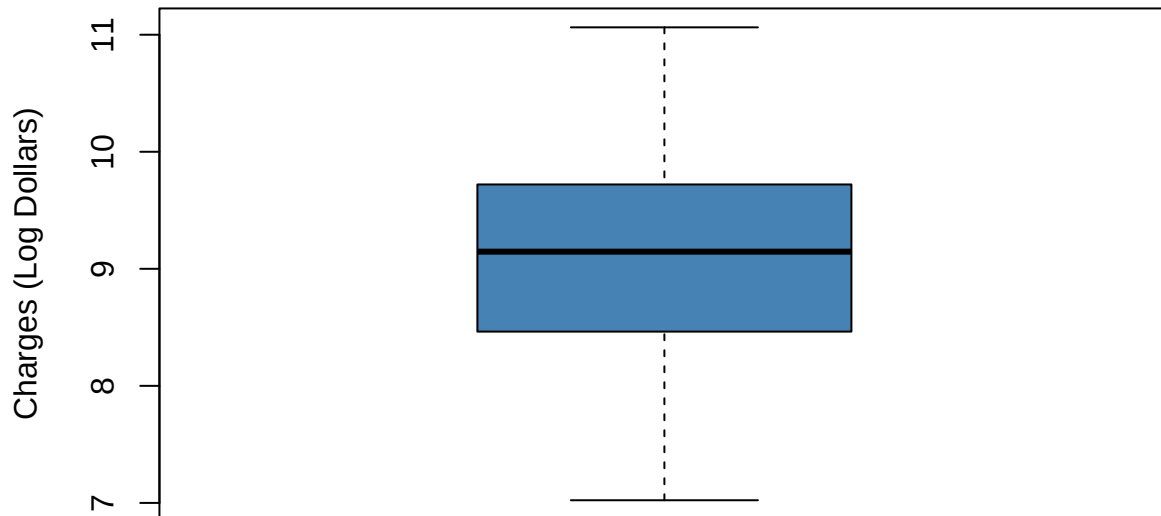
```
hist(df$log_charges, main="Histogram of Log Charges", xlim=range(7,12),  
      ylim=range(0,200), breaks=hist_breaks, xlab="Charges (Log Dollars)",  
      col="steelblue")
```



```
# Creates box plot of log charges
```

```
boxplot(df$log_charges, main="Box Plot of Log Charges",  
        ylab="Charges (Log Dollars)", col="steelblue")
```

Box Plot of Log Charges



```
# Sets x equal to charges
x <- df$charges

# Finds the first and third quartile of charges
Q1 <- quantile(x, 0.25)
Q3 <- quantile(x, 0.75)

# Gets the interquartile range
IQR_val <- IQR(x)

# Finds the lower and upper bounds
lower_bound <- Q1 - (1.5 * IQR_val)
upper_bound <- Q3 + (1.5 * IQR_val)

# Counts outliers
sum(x > upper_bound)
```

```
## [1] 139
```

```
# Counts number of males in data set
sum(df$sex == "male")
```

```
## [1] 676
```

```
# Counts number of females in data set
sum(df$sex == "female")
```

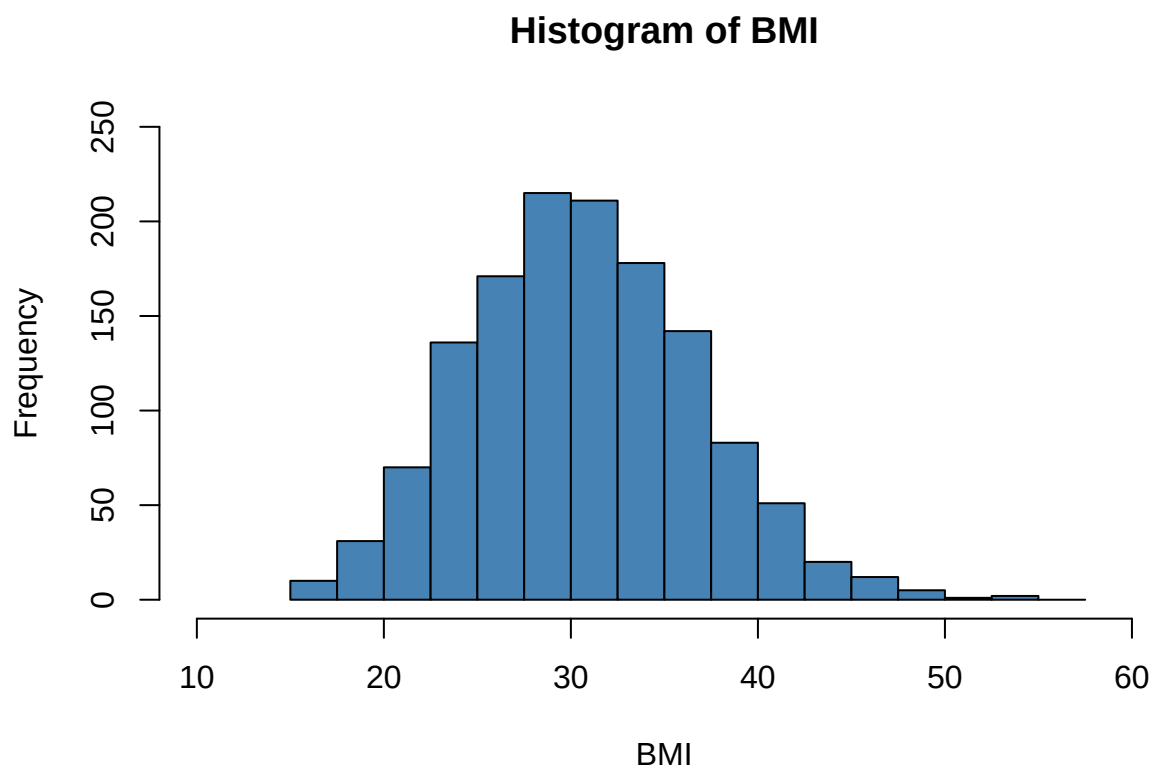
```
## [1] 662
```

```
# Makes breaks for histogram
```

```
hist_breaks <- seq(15, 57.5, by = 2.5)
```

```
# Creates histogram of BMI
```

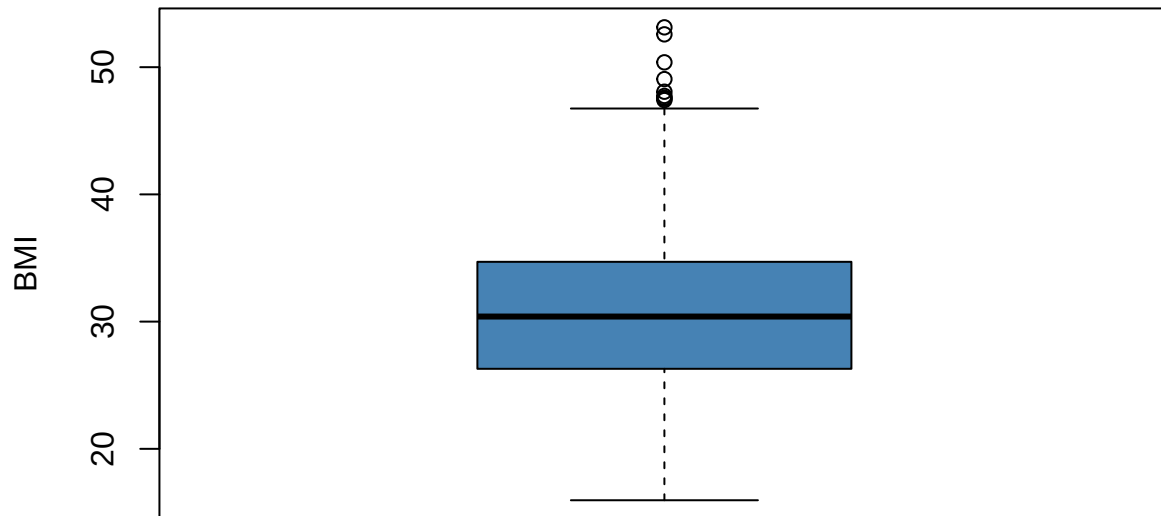
```
hist(df$bmi, main="Histogram of BMI", xlab="BMI", xlim=range(10,60),
     breaks=hist_breaks, ylim=range(0,250), col="steelblue")
```



```
# Creates box plot of bmi
```

```
boxplot(df$bmi, main="Box Plot of BMI", ylab="BMI", col="steelblue")
```

Box Plot of BMI



```
# Sets x equal to BMI
x <- df$bmi

# Finds the first and third quartile of BMI
Q1 <- quantile(x, 0.25)
Q3 <- quantile(x, 0.75)

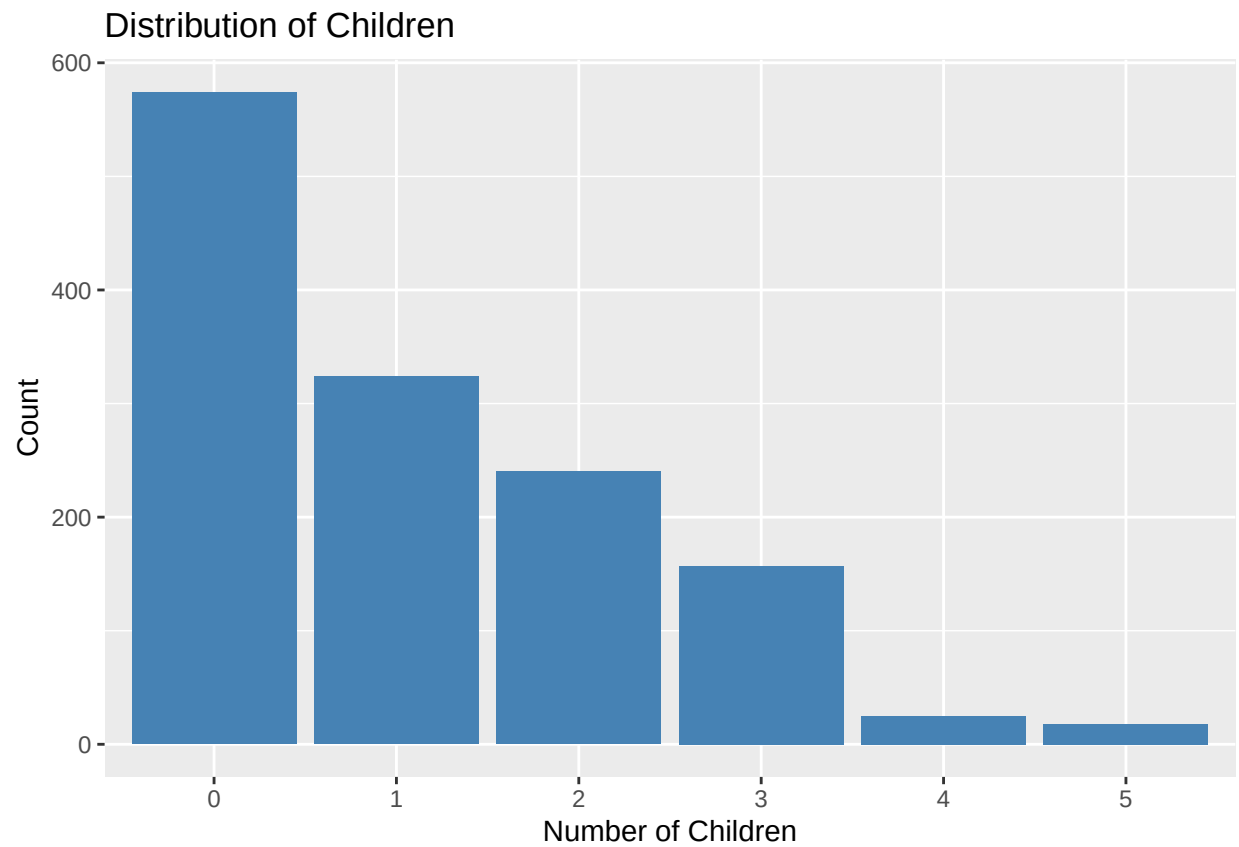
# Gets the interquartile range
IQR_val <- IQR(x)

# Finds the lower and upper bounds
lower_bound <- Q1 - (1.5 * IQR_val)
upper_bound <- Q3 + (1.5 * IQR_val)

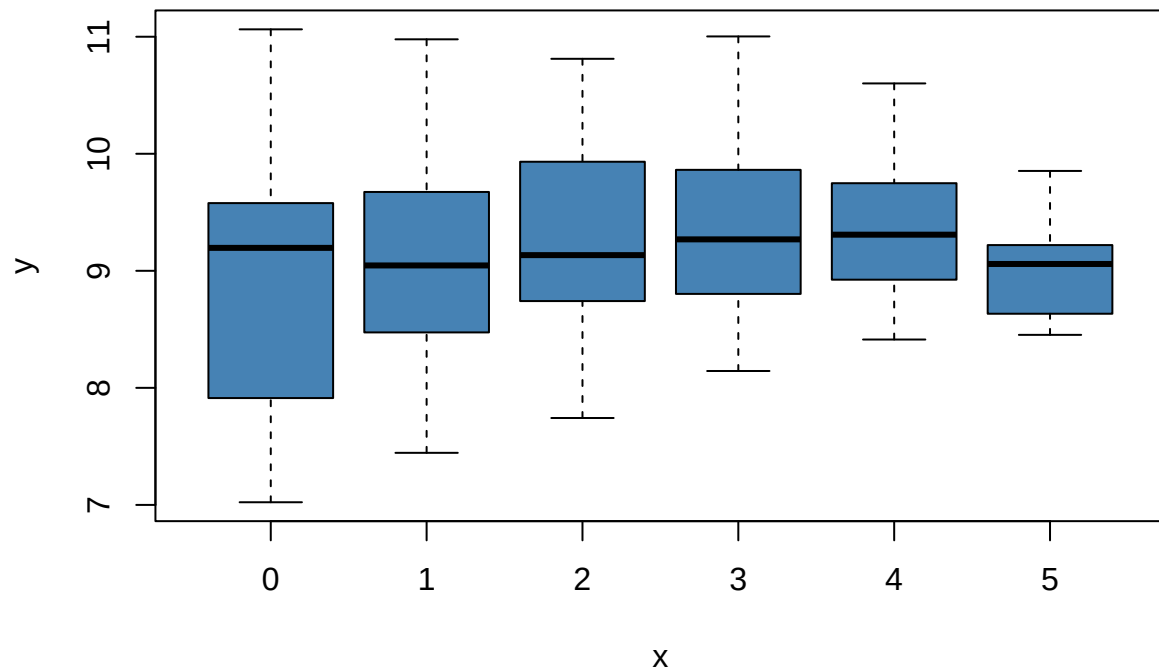
# Counts outliers
sum(x > upper_bound)
```

```
## [1] 9
```

```
# Creates bar plot of number of children
ggplot(df, aes(x = children)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Distribution of Children",
       x = "Number of Children",
       y = "Count") + scale_x_discrete(breaks = 0:5)
```

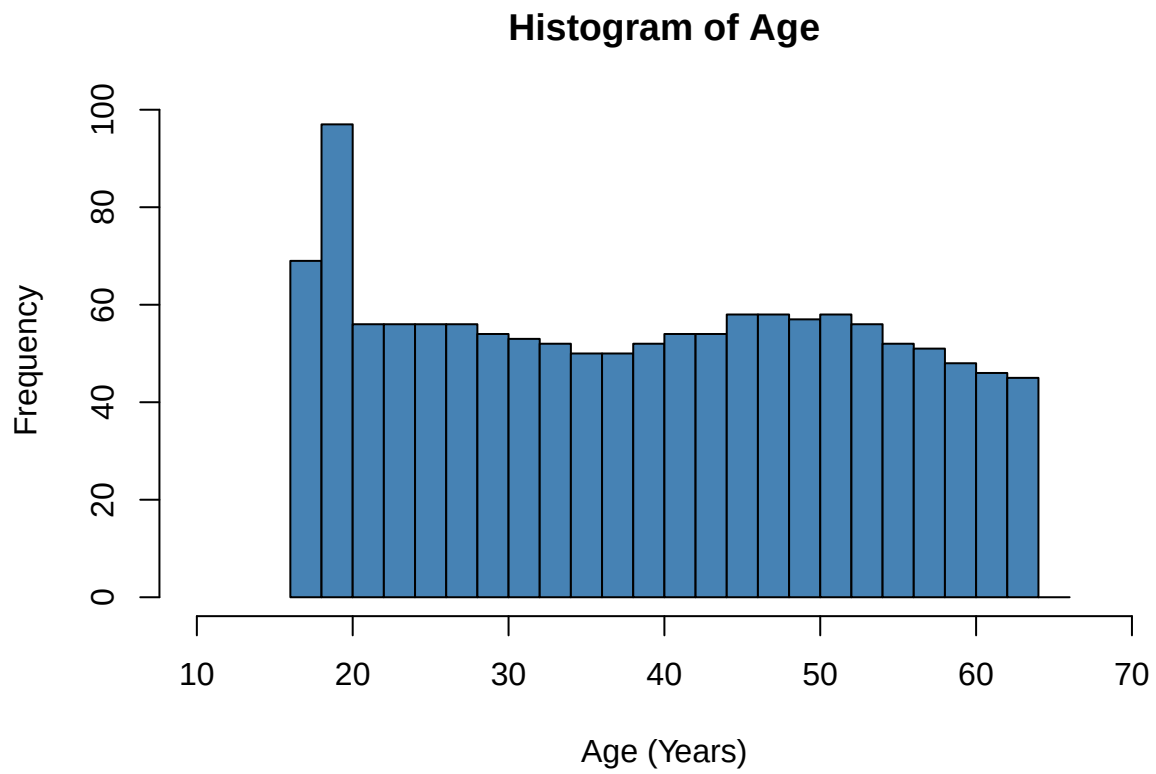


```
# Plots side by side box plots of log charges against number of children  
plot(df$children, df$log_charges, col="steelblue")
```

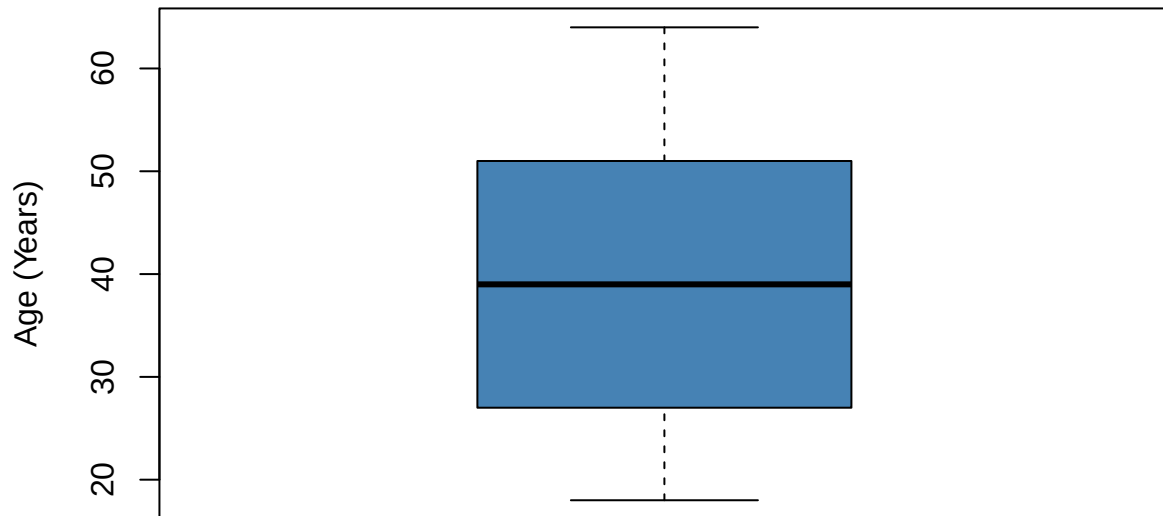
```
# Makes breaks for histogram
hist_breaks <- seq(16, 66, by = 2)

# Creates histogram of ages
hist(df$Age, main="Histogram of Age", xlab="Age (Years)", xlim=range(10,70),
     breaks=hist_breaks, col="steelblue")
```



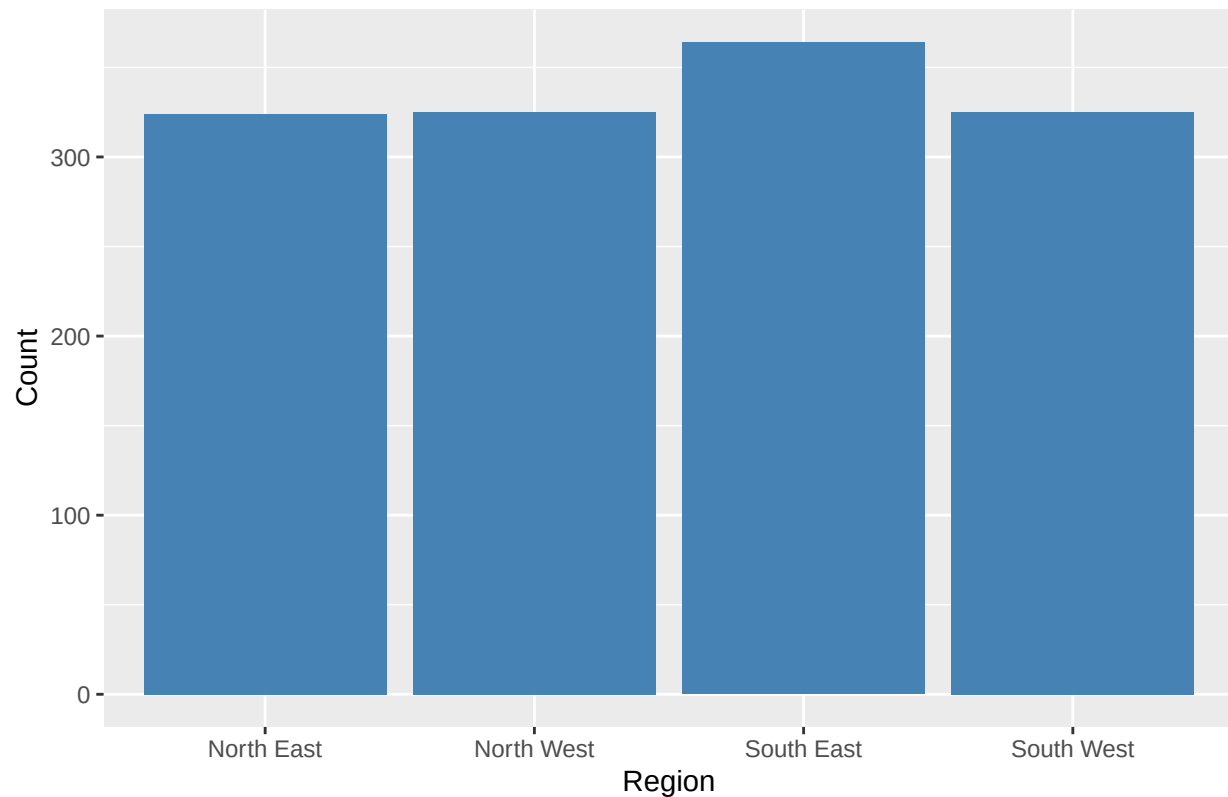
```
# Creates box plot of age  
boxplot(df$age, main="Box Plot of Age", ylab="Age (Years)", col="steelblue")
```

Box Plot of Age



```
# Shows bar plot of regions
ggplot(df, aes(x = region)) +
  geom_bar(fill = "steelblue") +
  labs(title = "Distribution of Region",
       x = "Region",
       y = "Count") + scale_x_discrete(labels = c("North East", "North West",
                                                "South East", "South West"))
```

Distribution of Region



```
# Counts number of smokers in the data set  
sum(df$smoker == "yes")
```

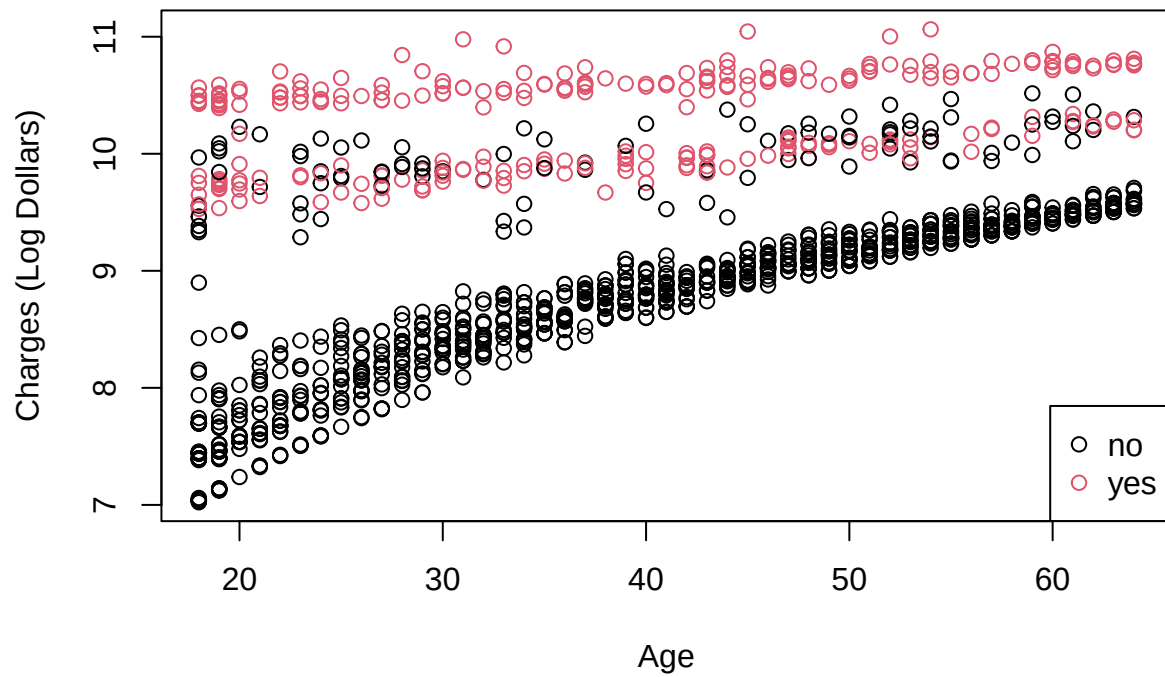
```
## [1] 274
```

```
# Counts number of non-smokers in the data set  
sum(df$smoker == "no")
```

```
## [1] 1064
```

```
# Plots age against log charges with smoking status overlay  
plot(df$age, df$log_charges, col = df$smoker,  
      main="Plot of Age and Log Charges Given Smoking Status", xlab="Age",  
      ylab="Charges (Log Dollars)")  
legend("bottomright", legend = levels(df$smoker),  
      col = c(1,2), pch = 1)
```

Plot of Age and Log Charges Given Smoking Status



```
# Gives the maximum charge for a male  
max(df[df$sex == "male",]$charges)
```

```
## [1] 62592.87
```

```
# Gives the maximum charge for a female  
max(df[df$sex == "female",]$charges)
```

```
## [1] 63770.43
```